

Electric Vehicle Adoption - A Comparison between Countries

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Abstract—This study covers an investigation of the adoption of EVs across the globe. The study makes use of Power BI which is a solid tool for doing visualizations that can be helpful in providing insights for decision making.

Index Terms—visualizations, EVs, PHEV, regions, BEV

I. INTRODUCTION

The global trends of EVs' sales have been increasing, and about 14 million new electric cars were registered in 2023 [1]. The key reason for adopting EVs is mainly to tackle rising environmental pollution, disuse of fossil fuel, and the effects of climate change [2]. In its endeavor to curb emissions of carbon, global leaders and organizations have developed thrilling goals on the shift of the utilization of traditional internal combustion engine automobiles to electric vehicles. The transportation sector is said to be spearheading the move to electric vehicles due to a study by the International Energy Agency IEA [3]. Introduction of electric vehicles as a concept is positive, but it comes with several problems as is highlighted below. Some of these barriers are charging infrastructure, range anxiety and the price of the electric vehicles are still major challenges, and most affected are those in the developing nations. However, there are other factors, which is cross-country/region perspective, that need to be taken into consideration in order to understand the issues related to the adoption of EV [5]. These challenges raise the question of appropriate approaches to data visualization that would allow policymakers, industry representatives, and the public to better recognize the dynamics and results of the transition to EVs at the international level [6].

A. Project Objectives

The purpose of this project is to develop an application that will show trends, barriers, and key factors for the electric vehicle market in the countries and regions throughout the World. The major aspects of visualization will be EVs sale, stock, and major powertrain type for improved analysis of EV penetration trend across the period and different regions. This project focuses on the adoption trend, and thus Power BI has been used to provide users with easy access to the data and create a visually friendly, easily filterable, and interactive dashboard where they can compare the trends collated by country and by year, as well as the relationship between EV adoption and a

reduction in carbon emissions. The idea is to show different countries' experiences on electric vehicles adoption, including the obstacles they encounter, like infrastructural conditions, consumer's awareness and policy measures. Finally, the work will help to identify recommendations that would be applied in policy-making and business development to foster global electrification of the automotive fleet.

B. Maintaining the Integrity of the Specifications

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C. Importance of Visualization

Global trends of electric vehicle (EV) adoption need to be visualized to embrace the future of sustainable transport. Because the transportation industry is one of the biggest culprits for greenhouse gas emissions, the use of electric cars serves as one of the most significant approaches to reversing climate change. But what has galvanized consumers to adopt it has been influenced by infrastructure, government policies, awareness and economic status of country to country basis. Thus, the project makes it possible for policymakers, industry stakeholders, and a broad range of other stakeholders to appreciate trends, potential difficulties, and necessary intercessions. It is suggested to use the basic forms of graphical representation of data, including bar graphs, tree maps, and time series to interpret complex data to decisionmakers to help them grasp the differences between regions, review the dynamics, and develop strategies on popularizing electric vehicles. It is therefore clear that effective visualizations are a key way of facilitating advocacy, policymaking and mass campaigns to bring about changes needed for the world to shift to environmentally sustainable means of transport.

II. BACKGROUND

A. Dataset Description

The data for this project is collected from various sources, however data mainly collected directly from the IEA and other government and industrial EV reports. This dataset offers historical and projected trends of electric vehicles' sales and stock, by regions and countries, from 2012 to 2035. The dataset is structured with the following key variables:

- **Region:** The geographical region or the country to which population data has been given. In the cases displayed, "Australia" is mentioned; however, other countries are present in the whole data array.
- **Category:** Depending on the type of data provided this field could be either 'Historical' or 'Projected'. The words "Historical" means actual sales or stock information existing about electric vehicles while 'Projected' can include forecasts about the EVs.
- **Parameter:** This simply refers to the exact measure under assessment. Key parameters include:
 - EV Sales Share: The Banking on EVs initiative defines the market shift as a ratio of the number of electric vehicles sold in a year to the total number of vehicles sold in that year and country/region
 - EV Stock Share: The proportion according to which share of total automobile fleet today is electric (existing number of vehicles).
 - EV Sales: Sales of electric vehicles within the year split by category (BEV for Battery Electric Vehicles for instance).
 - EV Stock: The stock of electric vehicles refer to the number of electric vehicles in use on the road today or today's stock preferably; they can be segmented based on any of the categories earlier mentioned such as BEV
- **Mode:** The make and model of the vehicle and even subbranches such as "Cars" in the given subset. Other vehicle modes include "trucks".
- **Powertrain:** This defines the category of electric vehicle powertrain and could be classified as EV (electric vehicles in general) or even BEV (Battery Electric Vehicles).
- **Year:** Year in which the reported data was extracted. The dataset is from the year 2011 up to the present and has an annual frequency.
- **Unit:** The measure unit that is used in the measurement of each parameter. In Verilog HDL this can be described by a simple statement involving units such as "percent," "vehicles," or with respect to a specific identified variable.
- **Value:** The real quantity associated with the unit for the identified year, area and characteristic.

For instance, according to the dataset, Australia sold 49 electric vehicles in 2011, corresponding to an EV sales share of 0.0065 % and an EV stock share of 0.00039% in the same year. They look at trends in using EVs in history with more detailed information on country like Australia, including sales and stock information over time [1].

B. Challenges in Visualization

Based on the gathered data, there are some challenges related to the visualization of the global EV adoption data: Data quality – The quality of the data varies significantly from country to country; Complete data – The complete datasets for completing the information are not available for all countries; Large scale Data – Working with large scale datasets is complex in general and especially when one is trying to identify meaningful insights from them. Indeed, one of the main issues when it comes to drawing a picture of the current and future use of EVs is the absence or lack of data. For instance, some nations are still either reluctant to release accurate data and statistics on EV sales or its inventory especially in the developing world. On the same note, different parameters of reports across the nations complicate this issue. Some countries offer subdivisions of different EV types, like BEV and PHEV while some only offer general numbers; thus, do not allow much comparison to be made [1]. The second problem notes the heterogeneity of the input data in different world regions. For instance, the USA, China or European countries have clear numbers of EV adoption, and upto-date datasets of high quality. However, other regions may use approximations, limited data or information, or even using outdated data. These disparities in data integrity must be corrected since well-sampled areas have the potential of skewing the overall results of visualizations if not adjusted for [2].

C. Existing Visualization Methods

The current and past years have seen several analyses and papers devoted to mapping out the advancement of electric vehicle (EV) ownership, employing diverse techniques to reflect adoption curves and disparities across geography and environmental outcomes. The IEA and other private research organizations have utilized different complex diagrams to analyze global EV data. These visualizations generally try to share trends of the rise in EV adoption, decline in carbon emissions, plus requirements for any electric vehicle. The best example of the use of data visualization in the field is the IEA's Global EV Outlook series of reports. To us, the IEA often employs line graph, bar chart, area plot to demonstrate the change of EV sales in different regions, with particular focus on the most developed EV markets which including China, EU and US [1]. For instance, in using the 2023 IEA report, stacked bar charts were utilized to show the changes in the market share of BEVs and PHEVs over time. These diagrams help readers to get an idea about the rate of adoption in different areas, and they also help to compare the countries that differ in basic developments and state incentives. Similar to the mentioned methods, heat maps and choropleth maps also illustrate data on the presence of electric vehicles by regions. Such maps are used in research like 'Global Adoption of Electric Vehicles' by Anderson et al, (2022) [2] to represent electric vehicle markets by regions and countries with some grouping the countries by market segment sharing percentages into different levels. These maps are also effective in visually illustrating

geographic difference and can easily identify countries of most improvement or difficulties. Despite this heat maps can be useful at times and may offer a myopic view of the level of EV adoption as illustrated by the adoption percentages above without regard to charging facilities and government policies. Apart from these more conventional approaches, there have been some attempts to engage interactive visualizations. Features such as dynamic dashboards in Microsoft Power BI version and tangible Tableau, enable users to discover more about the adoption or frustration of EV. For example, the EV Adoption Visualization Tool designed by the National Renewable Energy Laboratory (NREL) shows the trends of EV adoption, charging stations, and emission reductions for selected location and vehicle type [3]. This tool is also useful to use the sliders and filters in order to get more detailed information and to filter data by users' preferences.

III. VISUALIZATION TECHNIQUES AND ETHICAL FRAMEWORK

A. Techniques Used

In this project, specific use was made of Power BI to develop visually engaging and interactive dashboards to explore more about the adoption of electric vehicles (EVs). The available tools in Power BI such as the multi-types of chart and filter, interactive options all provided a strong support for presenting the dataset and dealing with cross-country and cross-regional comparison.

1) *Visualizations*: The primary technique for visualizing EV adoption trends involved the use of several graphical representations designed to present the data clearly and effectively:

- **Bar Graphs**: Different measurement units were also counted within the dataset and bar graphs were used to illustrate the occurrence of each of them including GWh, vehicles, percent, and so on to facilitate relative comparison of the measurement units.
- **Pie Charts**: To provide a proportional representation of EV adoption by different powertrains like BEV, PHEV, FCEV, or EV, pie charts were used. These charts visually depicted the percentage contribution of each powertrain, offering a quick snapshot of the dominant ones and their relative shares within the dataset.
- **Donut Charts**: Similar to pie charts but with a more aesthetically modern design, donut charts were employed to visualize data categories. The central space in donut charts allowed for additional information or key statistics, enhancing interpretability while still maintaining clarity of proportions.
- **Filled Maps**: For a more immersive geographical representation, filled maps were created. These maps used varying colors to represent levels of EV adoption across different regions.
- **Area Charts**: To illustrate changes in EV adoption over time and their cumulative effect, area charts were used. These charts emphasized trends and growth, showing not only the rate of adoption over the historical periods but also highlighting the projection-APS period as well.

- **Line Charts**: Trends in EV adoption across time were also represented using line charts. These charts offered a clear depiction of growth patterns, seasonal variations, or dips in adoption rates, providing an analytical tool for identifying long-term trends and forecasting future growth trajectories.
- **Filters**: Facilitate interactive data exploration by region, year, and powertrain type.

B. Ethical Framework

It is imperative to be honest and ethical while representing the global EV adoption data in the course of visualizing them. There are seven possible biases that can skew the way that the data can be displayed, and these have to be dealt with carefully as their presence may make the users of the visualizations misled or the crucial factors are hidden from plain sight. This section will also explain the primary ethical concerns and how these are addresses.

1) *Data Bias and Incompleteness*: Perhaps the only issue is a lack of complete data as an impediment in visualizing the EV adoption from across the world. As explained earlier, some nations provide information on EV usage more frequently and accurately than others do. For example, many developing nations either do not have specific figures regarding the number of EVs sold or the stock of these vehicles, or, in the best-case scenario, supply fragmentary information because the generally weaker state of statistics in these countries. To address this, we were keen to state the limitations of the dataset used in the creation of the visualizations before the explanations were given. The disclosure of missing or incomplete data as a factor in the visualization is crucial to helping the user see how incomplete data can affect the graphics, for example, where countries with no data are presented as insets.

2) *Representation Bias*: Another type of bias is likely to arise where visualizations give undue prominence to certain areas, due to either the ability to obtain or the method of presenting the data. For instance, the major players in overall EV utilization are commonly the China, the United States, or European countries, so one can hardly have the expression that EV utilization is predominantly Western or Asian business. To mitigate this representation bias, we used proportional representations in creating our graphs and maps, and whenever possible, we sought to feature countries that were relatively small in number but witnessed emerging trends of adoption. It also helps users to have the big picture in mind while specific icons may not make smaller yet developing countries EV markets seem important.

3) *Clarity and Simplicity*: Ethical data visualization also speaks to clarity and accuracy of the visualized information. Our goal when creating the Power BI dashboard was to create charts, maps, graphs that not only look good but are also easy to read. We decided against adding various elements which unnecessarily complicate the label or contain too many colors, which only perplex the user or hide important information. The layout of the visualizations was meant to stress the primary trends of interest – the rates of Electric

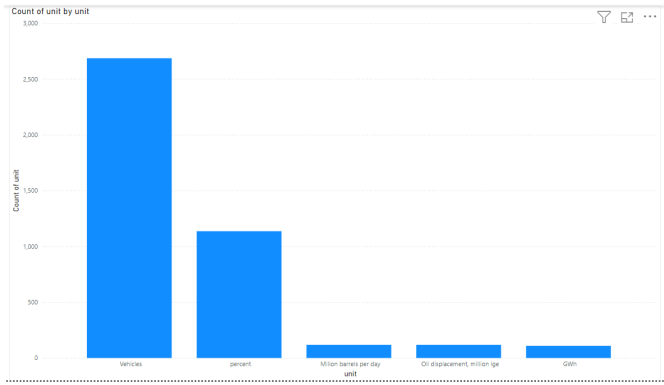


Fig. 1. Unit Frequency Bar Chart

Vehicle adoption, emissions reductions, and some country-level comparisons—while permitting users to interact with the visualizations more flexibly and to filter insights, seamless.

IV. PERSONAL APPROACH

A. Chosen Visualization Methods

In this project, Bar graphs, pie charts, donut charts, filled maps, area charts, line charts, and filters were selected for their effectiveness in presenting comparative data. These methods provide an intuitive overview of global EV adoption trends.

B. Justification of Choices

Among the selected contribution categories, bar graphs, tree maps, and filters were used because they allowed presenting data on the implementation of electric vehicles in different areas and at different stages of development. These methods do not only help to enhance the visibility of the data but also helps to make the interactions with the data on the part of the user richer, thereby enabling the user gain a right perspective based on the trends and reverses of EV adoption all over the world.

C. Screenshots & Descriptions

In this section, more detailed descriptions of the visualizations developed for the EV adoption project are described. The trends found in EVs were analysed and presented for different regions and powertrains in the Power BI and these are presented below. The following is a description of the eight primary indicators of visual learning, and observations from the findings of the study.

1) *Unit and Measure Frequency Graph:* The first graph is a bar graph that categorizes the common units and measures used in the dataset. This graph is useful in showing how frequently various units recur within the data set; it therefore gives an idea of the measurement forms employed to capture EV uptake.

Insights: The most frequently occurring unit in the data set is “vehicles;” it is mentioned 2,487 times, which, most probably, means the number of electric vehicles sold or in stock. The second most often used measurement is “percent” which is found 1,068 times probably as the percentage of the

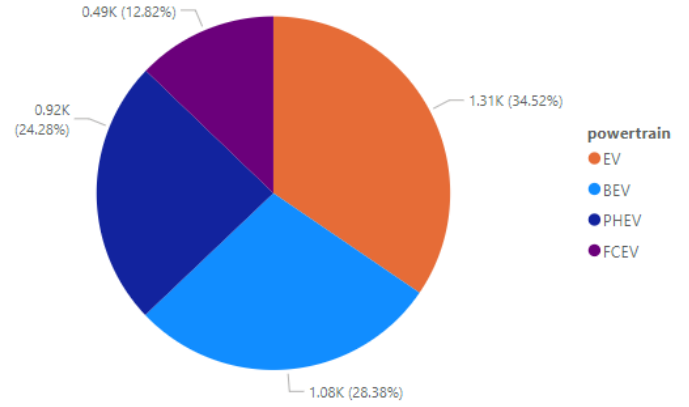


Fig. 2. EV Cars based on powertrain (Historical)

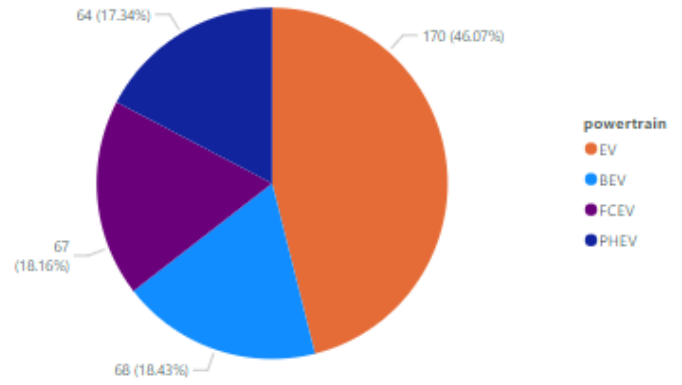


Fig. 3. EV Trucks based on powertrain (Projected)

market share or the percentage increase in the use of EVs. Terms such as “Millions barrels per day” and “GWh” were used 84 and 75, respectively; terms associated with energy and oil consumption. This graph is also useful to bring attention to what sorts of metrics are being used and how frequently they appear in the data.

2) *Average Value by Powertrain Mode:* The second pie charts represents the EV based on powertrain modes where you have the mean number of electric cars in each category (BEV, EV, FCEV, and PHEV). They have also provided a preferable form to illustrate the regional classification of EV type and the share of every category in the total EV demand. Overall amount of Electric vehicles are derived as below:

- BEV (Battery Electric Vehicles): 297 Million
- EV (Electric Vehicles): 131.71 Million
- FCEV (Fuel Cell Electric Vehicles): 4.77 Million
- PHEV (Plug-in Hybrid Electric Vehicles): 2.4 Million

Insights: This to some extent suggests that the dataset is all about the stock and penetrations of electric vehicles, as

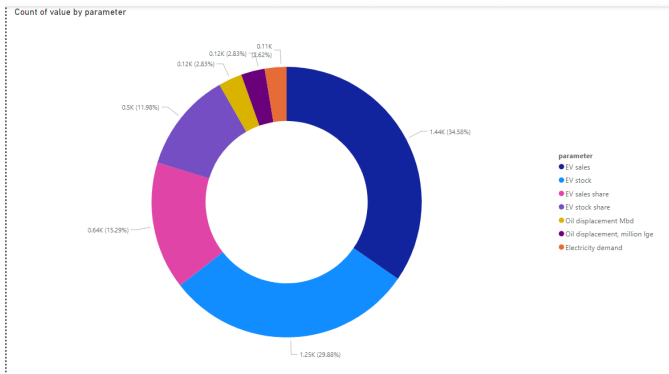


Fig. 4. Average Value by Parameters

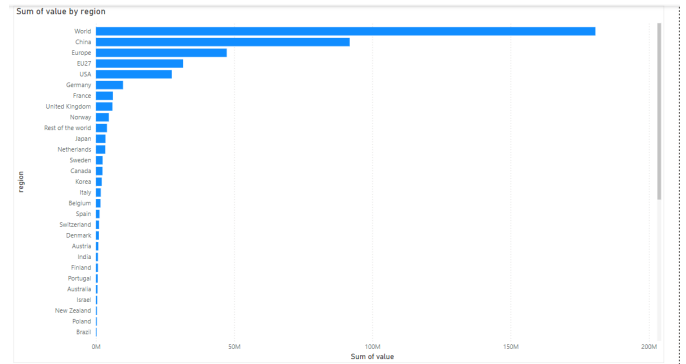


Fig. 6. Distribution by Region (filtered by Value)

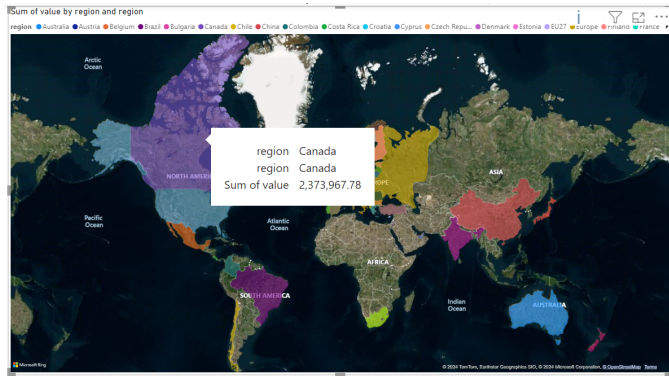


Fig. 5. Distribution by Region based on Adoption Values

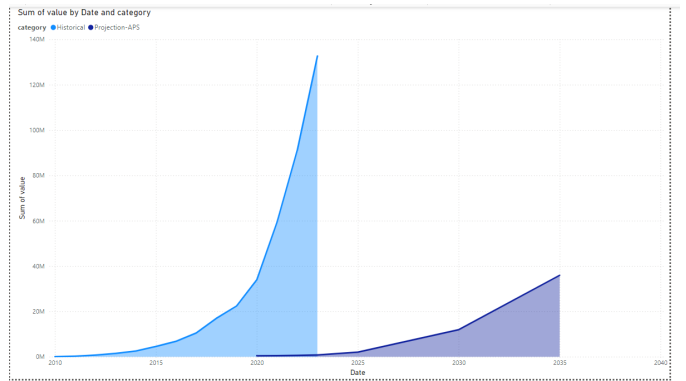


Fig. 7. Yearly Trends in EV Adoption - filtered by Category

reflected by the predominance of ‘vehicles’ and ‘%’. The frequency of energy-associated metrics (GWh, million bpd) is somewhat lower than that of vehicles or market share, which still underlines the fact that although energy consumption is metered and comparable, it is not a priority as the quantity of automobiles and market share.

3) *Average Value by Parameter*: The third donut graph represents the mean by different parameters where you have the mean number of electric cars based EV Stock, EV Sales, Electricity Demand, Oil Displacement, EV Stock share, EV Sales share, etc.

Insights: The largest segment, “EV Stock,” accounts for 74.07%, dominating the data. “EV Sales” follows at 25.36%, indicating significant but smaller contributions. Other parameters, such as “Electricity Demand” and “Oil Displacement,” are minor, reflecting a minimal share. This highlights the focus on growing EV stock and sales while other metrics contribute marginally.

4) *Distribution by Region (filtered by Value)*: These graphs explores the regional distribution by powertrain type of EVs. This visual offers a geographic perspective of EV uptake and is an answer to the question of which regions are ahead in different powertrain types. If the filter is set for that specific powertrain, the graph clearly shows that the distribution of EVs and the presence of BEVs, PHEVs and other powertrains is not equal across the regions.

5) *Yearly Trends in EV Adoption*: We can see yearly trends in the EV adoption process through time series graphs. There was consistently increasing number of EVs sold and a regular increase in market share over the years, which is indicative of EV adoption on the rise. The data, however, shows a time delay of the EV adoption in developing countries, where EVs have been slower to catch on, owing to such stumbling blocks as infrastructure constraints and economic determinants. Such insights call for targeted policy and intervention to push electric vehicles into emerging markets.

D. Final Dashboard

Fig 7 shows the overall dashboard we have made. On the top left we have the pie chart that depicts the distribution of several types of EV powertrains, including battery electric vehicles (BEV), plug-in hybrid electric vehicles (PHEV), fuel cell electric vehicles (FCEV), and generic EVs. BEVs lead the market, accounting for a huge 68.13% share, followed by PHEVs (30.21%). The other sectors, which include FCEVs and other EVs, make up a small proportion. This animation highlights the market’s preference for completely electric automobiles over hybrid and fuel cell technology.

The gauge chart compares the aggregated EV value, which is around 436.01 million units, to a benchmark of 872.03 million units. This statistic provides a high-level overview of the market’s progress toward a certain target or prediction. It

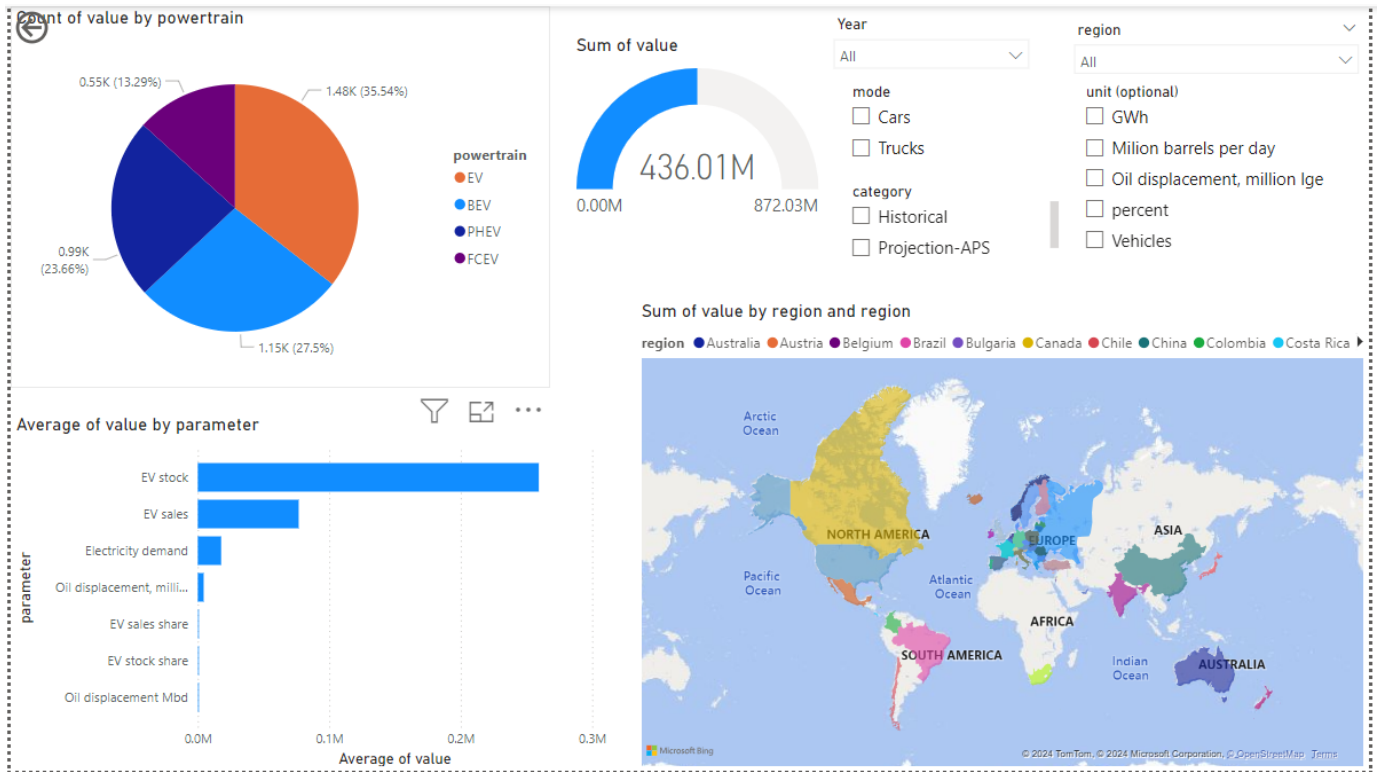


Fig. 8. EV Adoption Dashboard

is a wonderful tool for tracking worldwide EV adoption over time or evaluating achievement against predefined goals. The bar chart categorizes several EV-related parameter, including EV stock, EV sales, power demand, EV sales share, and oil displacement measures. The graphic shows that EV stock has the highest value, followed by EV sales, showing a significant increase in the number of EVs on the road. Other indicators, such as power demand and oil displacement, indicate the environmental and economic implications of EV adoption. The granularity of this figure is useful for stakeholders evaluating contributions to sustainability and energy transition goals.

The filled map offers a geographic view on EV measures, including data from North America, Europe, Asia, and South America. Each region is color-coded and labeled with the relevant data points, allowing visitors to see how various places contribute to global EV trends. Countries like as China and the United States, which are well-known leaders in EV adoption, may be evaluated to determine their relative shares. The map also allows for comparisons between developed and emerging regions in terms of EV stock, sales, and other variables.

The dashboard includes slicers for key variables including year, mode (cars or trucks), category (historical or predicted statistics), and unit type. These slicers allow users to engage interactively with the visuals, adapting the analysis to their individual needs. For example, selecting "Historical" under the category allows users to study previous patterns, whilst selecting "Projection-APS" gives insights into future possibilities under advanced policy settings.

E. Data Insights:

What stands out as one of the biggest insights from the visualizations is the variable rate of EV adoption across regions. Bar graphs and filled maps indicate very clearly where countries such as China and the United States are leading the charge in EV adoption, always with the highest values in vehicle stock and market share. As an example, the bar graph of the "distribution by region": China and USA has highest sales for EV category, meaning those countries have a high value for EV adoption as well. However, in Europe, countries like Norway, Sweden and The United Kingdom also have strong adoption, being as high as close to 28 vehicles per 1,000 people (e.g., Norway). It indicates that European countries, namely those that implement progressive environmental policies have played a key role in the global EV market. But the bar chart visualization still shows that most regions with high adoption rates take up a small fraction of the global market and that a few key players dominate the larger markets. The thesis is thus reinforced: while global EV adoption is rising, the roll out is not uniformly spatialized, with some regions enjoying an early lead. Moreover, the distribution by powertrain visualization further reveals what kinds of powertrains are adopted in what area. For instance, the leading regions (in terms of EV adoption)—China, USA and Europe—are mostly following the path of BEVs (Battery Electric Vehicles), with PHEVs (Plug-in Hybrid Electric Vehicle) adoption spread across regions equally. The reliance on hybridised powertrains however indicates that even in countries with strong BEV

uptake, hybrid technology is broadly present and popular. And this means that there is no one size fits all solution and different regions are making different choices about how to cut emissions and adopt clean energy vehicles.

The averages of BEVs are, on balance, much higher than all the other powertrains, mainly because BEVs account for a much greater proportion of electric cars on today's market. There are so many BEVs than FCEVs or PHEVs, and since the total average value of EV and PHEV is lower than that of BEV, they are smaller powertrain categories. This revealed that the proportion of battery electric vehicles (BEVs) has increased as the leading type of EVs around the world.

Polymakers, automakers, and sustainability experts can all benefit greatly from the final dashboard's strong decision-support capabilities. It enables focused interventions and investments by clearly illustrating geographical inequalities, powertrain choices, and EV adoption patterns. The focus on factors like oil displacement and power demand highlights the wider ramifications of the EV transition for energy systems and climate objectives.

F. Challenges in Global Adoption

Finally, the visualizations expose the problems that still remain in actually achieving extensive EV adoption. To illustrate, adopting regions still experience limitations in reaching the next level of scaling the market, including lack of infrastructure support for charging stations, low public awareness, and vehicle cost. In countries where EV adoption is lower, there may not be sufficient incentives, or government support, to make them a consumer option. Understanding the dynamics of EV market, these insights have importance in order to overcome these barriers, and you can see the importance of data driven policy.

V. CHALLENGES AND REBUTTAL

A. Implementation Challenges

As part of this project, there were a number of technical and data related challenges with which we were able to create the visualizations, however planning and improving was an iterative process and we spent the time troubleshooting to get the information correct. We covered data cleaning, integration, visualization challenges, and had to make sure we could be interactive with those data. Below is a review of the most crucial issues present in the implementation phase and their solutions.

1) *Data Quality and Cleaning:* Another of the first issues was trying to ensure the quality and cleanliness of the data. The data also had some inconsistencies (missing values, duplicate entries, and mismatched units of measurement, as is typical with any large real world dataset). For example, there were records with missing values for key metrics such as "EV sales share" or "EV stock share", whose accurate visualization is important. More troublesome for countries with very few EV registrations or very early years in the dataset were the missing data points.

2) *Data Integration across several sources:* This project used a dataset that was spread across multiple sources and each source had a different format, making things more complicated. Some of the data came in as .csv files, and some data was in proprietary formats. Regional data was separated into individual files for every state in the United States, and global data was lumped into one final file. Accuracy of the comparisons between countries required these data formats to be integrated and categories and units to be set consistently.

3) *Handling Large Datasets:* Handling the large volume of data was another great challenge. The dataset contained long spans of years (2011 through 2023) for thousands of countries which resulted in many rows and columns. However, this amount of data was increasingly difficult to visualize efficiently, let alone in ways that are userfriendly, in Power BI in particular, as it can slow down and appear unresponsive when dealing with such large datasets.

4) *Making It Interactive and Usable:* A key goal of this project was to make interactive visualizations for the user to be able to simply explore the data themselves. To enable this, it meant bringing in favourites, slicers, and drillthrough features to Power BI to offer users the ability of looking at what data they want to consider. But people tasked with creating these interactive elements needed to make sure they were both functional and user friendly.

5) *Design and Aesthetic Challenges concern:* A further challenge was to piece together a coherent design with a beautiful aesthetic while the visualizations were as functional and virtually interpretable as possible. There was simply too much data and we needed to balance detail and clarity. If there is too much information on a graph, it will overburden the consumer, if there is too little the depth of the analysis will be undermined.

B. Critique & Feedback

The data for the visualizations were developed iteratively with valuable feedback received from my group member to improve the final results. First, the visualizations served to show the top line trends and data points, but after a quick discussion with my group member it became apparent that there was room for improving the user experience to improve the clarity and interactivity. The visualizations may not offer enough detailed comparison between countries or regions that could provide deeper insights, such as comparing countries with similar EV adoption rates or highlighting trends that diverge significantly. Among the most important feedback included filtering options in the visualizations. My group member also suggested that adding another level of filters to make data more interactive and explore more dynamically like choosing specific countries, years or powertrain types would help the users. Based on the feedback from these analytics, I added more filters to Power BI so that the users can select a different region, powertrain, and time frame to better tailor the visualizations to users' preferences. The completely re-written dashboard has drastically improved the interactivity, including the ability to drill down to specific subsets of the data and

to compare trends between countries. Although adding filters for interactivity was a good move, the overall UX might still be overwhelming for non-expert users who are not familiar with data visualization tools like Power BI. There was another critique on the tree maps, which looked murky first glance. One of my group members said that the color coding and size of the rectangles might not be intuitive for people who don't know anything about hierarchical visualizations. To account for this, I modified the color schemes and added tool tips with further information so that the map would be easier to understand. This referenced tree maps revision improved the tree maps overall usability and readability. While the tree maps have been improved, other elements such as charts, color schemes, and labeling across different visualizations may lack consistency. Inconsistent design choices can confuse users.

C. Rebuttal

It is an honor for the feedback I received during development of the visualizations as it really helped refine the visualizations and make it more interactive and user friendly. we nonetheless insist on the final visualization method, and believe it is a highly effective visualization method for this project's goals. By combining different charts and filters, we represent complex EV adoption trends across regions and time in a comprehensive and intuitive way. Stacked bar charts for BEV compared to PHEV, for example, let users quickly discern how the market is moving. Despite initial criticisms, tree maps provide a distinct and easily digestible form of presentation for the relative size of each country's market share in a hierarchical structure that is needed to understand global trends on a compact scale. Filtering was chosen to make it more interactive and it was done on purpose. In a global context, where many users can come with different interests (are interested in specific countries or points in time), dynamic data exploration relies on being able to explore subsets of the data that follows users' interests. With feedback, I believe the visualizations, first, very effectively convey key insights, and secondly, it creates an engaging and user centric experience.

VI. CONCLUSION

A. Summary of Findings

The generated visualizations for this project capture key takeaways about the global landscape of electric vehicle (EV) adoption. The project looks at countries' trends in EV sales, stock, and powertrain distribution and finds that countries like the US and China are leading the charge on EV adoption compared to Europe. The bar graphs and tree maps demonstrate how large a slice of the market these regions have occupied, and smaller markets in Europe, the UK and Australia are also seeing significant progress. Additionally, powertrain type analysis shows different adoption patterns between BEVs and PHEVs, with BEVs being more adopted in certain regions and PHEVs being more adopted in all countries. The importance of regional policies, infrastructure, consumer preferences and the risks associated with the fade out value are emphasized by these findings.

B. Future Directions

The current visualizations offer useful insights, however future work could examine the role government policies and EV incentives play in impacting EV adoption rates in emerging economies. We may also want to make socio economic factors such as income levels, urbanization, and charging infrastructure part of our understanding of the EV adoption trend. Other work in the future could also include expanding the dataset to other recent years as well as additional EV related metrics like environmental impact and lifecycle assessments of electric vehicles.

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